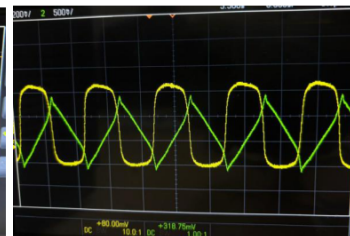
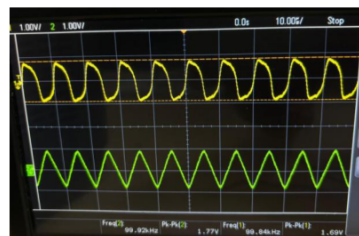
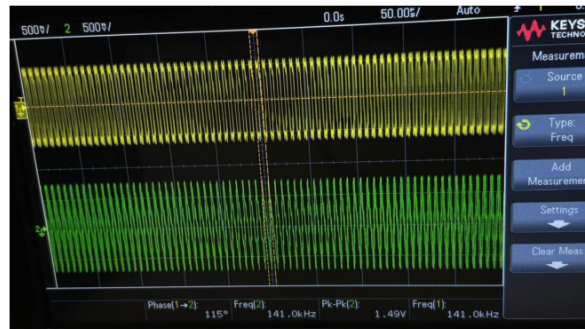


Quadrature Down Converter (QDC)

Quadrature Oscillator

Quadrature Down Converter (QDC) – Images from DSO

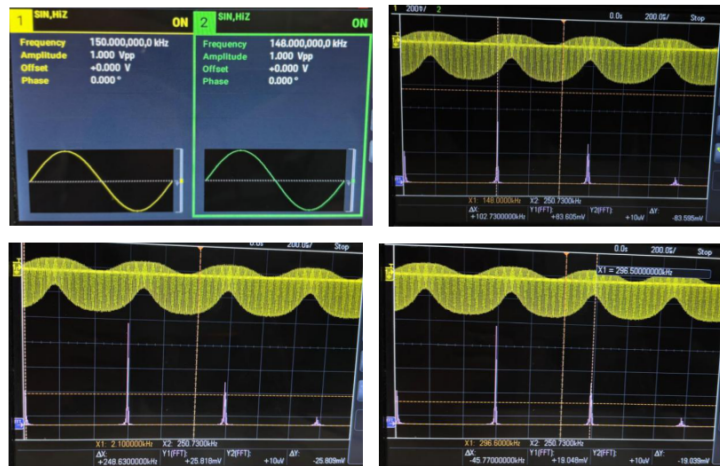
QUADRATURE OSCILLATOR



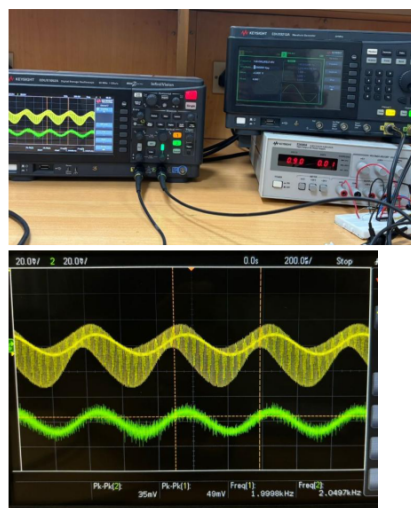
Waveform captures from the quadrature oscillator showing phase-shifted outputs and stable oscillation characteristics measured using the DSO.

Mixer Circuit and Low-Pass Filter

MIXER CIRCUIT



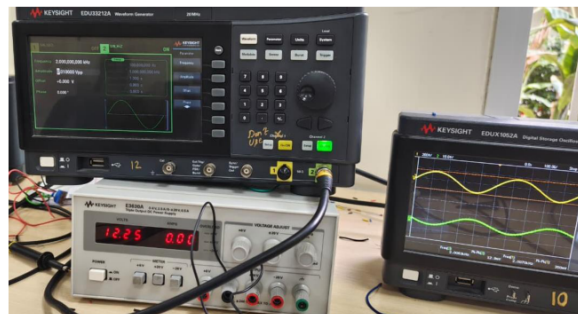
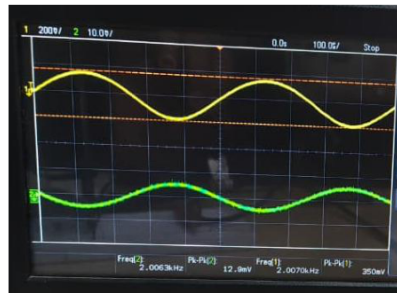
LOW-PASS FILTER



Experimental observations from the mixer circuit and low-pass filter stages validating frequency conversion and signal filtering performance.

Amplifier Stage

AMPLIFIER



Amplifier stage output measurements demonstrating signal amplification and waveform integrity during hardware testing.